

AUTOSAR High-Level Design: Vehicle Control Unit (VCU)

Electric Vehicle Powertrain Control | Revision: 1.0

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AUTOSAR Release	Classic Platform 4.4.0	Safety Level	ISO 26262 ASIL-D
Subsystem	EV Powertrain & Torque Mgmt	Revision	Version 1.0

1. System Overview and Torque Coordination

The Vehicle Control Unit (VCU) serves as the master supervisory controller for an electric vehicle powertrain. In revision 1.0, the architecture implements multi-core AUTOSAR Classic mapping on Infineon AURIX TC397. The VCU performs driver pedal interpretation, traction torque calculation, regenerative braking blending, and battery state-of-charge (SoC) power derating.

2. SWC Architecture & Allocation

SWC Name	Type	Periodicity	Core Allocation	Role
SWC_TorqueCoordinator	ApplicationSWC	10 ms	Core 0	Interprets accelerator pedal position and computes drive torque demand.
SWC_RegenBrakeManager	ApplicationSWC	10 ms	Core 0	Blends hydraulic friction and electric motor regenerative deceleration.
SWC_ThermalSupervisor	ApplicationSWC	100 ms	Core 1	Manages coolant pump and radiator fan speeds based on battery/inverter temps.

3. Inter-SWC and Inverter Interfaces

Communication with the Traction Inverter occurs over an isolated CAN FD private bus (5 Mbps baud rate) via AUTOSAR COM and CanTp. Torque requests are governed by interface *If_MotorTorqueCommand* with a freshness counter and end-to-end (E2E) Profile 4 protection.